

TCX2101: Calculus and Linear Algebra for Computing

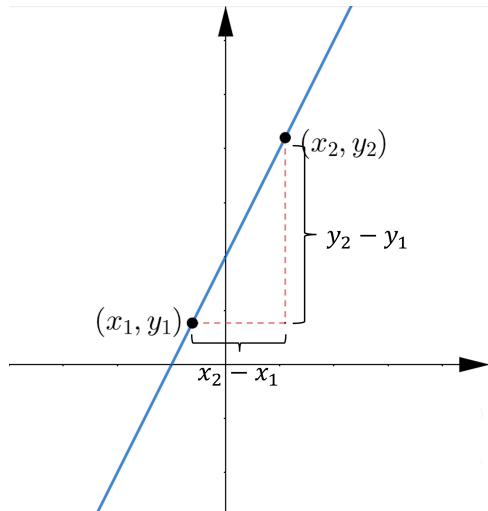
Chapter 3: Differentiation and Applications

Tangent Lines and Derivative

Example

The **gradient** of the **linear function**
 $y = mx + c$,

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$



Example

In general, the **gradient** to the curve $y = f(x)$ is approximately

$$\text{Gradient} \approx \frac{\Delta y}{\Delta x}.$$

The **tangent line** to the curve $y = f(x)$ at $P(a, f(a))$ is the line that pass through P with the above gradient.

<https://www.geogebra.org/m/mcxsfykc>

Remark

Required to take **limits** to find the gradient.

Definition

Definition

Suppose f is a function defined on an **open interval** containing a . The **derivative** of f at a , denoted as $f'(a)$ is the **limit**

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a},$$

provided the **limit exists**. The **tangent line** of the curve at $P(a, f(a))$ is

$$y = f(a) + f'(a)(x - a).$$

Remark

- (i) The derivative of f at a can also be denoted as $\left. \frac{d}{dx} \right|_{x=a} f(x)$, or $\frac{d}{dx} f(a)$.
- (ii) If $f'(a)$ exists, we say that f is **differentiable** at a .

Example

Let

$$f(x) = 3x - 2.$$

Find the derivative of f at 1, and the tangent line of the curve at the point $P(1, 1)$.

$$f'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{(3x - 2) - (1)}{x - 1} = \lim_{x \rightarrow 1} \frac{3(x - 1)}{x - 1} = 3.$$

So, the tangent line is

$$y = 1 + 3(x - 1) = 3x - 2 = f(x).$$

Example

Let

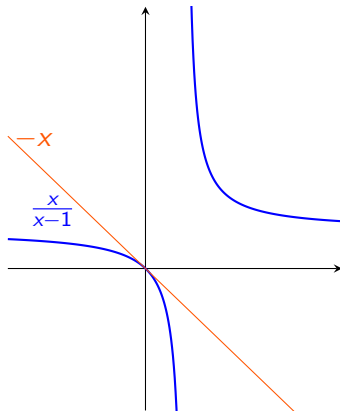
$$f(x) = \frac{x}{x-1}.$$

Find the derivative of f at 0, and the tangent line of the curve at the point $P(0,0)$.

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{h}{h-1} - 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h-1} \\ &= -1 \end{aligned}$$

So, the tangent line is

$$y = -x.$$



Example

Find the derivative of

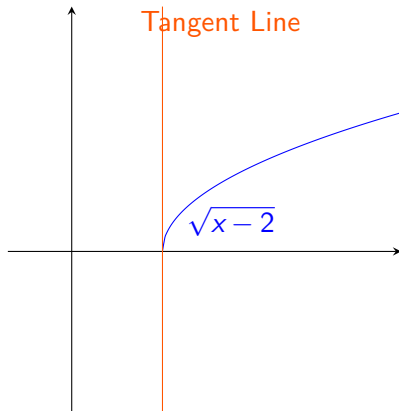
$$f(x) = \sqrt{x-2}$$

at $x = 2$.

$$\begin{aligned} f'(2) &= \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{\sqrt{x-2} - 0}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{1}{\sqrt{x-2}} \end{aligned}$$

does not exist!

To be precise, we should take $x \rightarrow 2^+$. This is called the right derivative of f at 2, see later.



Example

Find the derivative of

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

at $x = 0$.

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^2 \sin\left(\frac{1}{h}\right) - 0}{h} = \lim_{h \rightarrow 0} h \sin\left(\frac{1}{h}\right)$$

Since $-|h| \leq h \sin\left(\frac{1}{h}\right) \leq |h|$, by squeeze theorem,

$$f'(0) = \lim_{h \rightarrow 0} h \sin\left(\frac{1}{h}\right) = 0.$$

Derivative as a Function

Let

$$f(x) = \frac{1}{x}, \quad x \neq 0.$$

For any value $a \neq 0$,

$$\begin{aligned} f'(a) &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a} = \lim_{x \rightarrow a} \frac{\frac{a - x}{xa}}{x - a} = \lim_{x \rightarrow a} \frac{a - x}{xa} \frac{1}{x - a} = \lim_{x \rightarrow a} \frac{-1}{xa} \\ &= -\frac{1}{a^2} \end{aligned}$$

We can define a function

$$a \mapsto f'(a) = \frac{-1}{a^2}, \quad a \neq 0.$$

We call this the derivative of f .

Differentiable on an Interval

Definition

A function f is **differentiable** on an **open interval** (a, b) if it is differentiable at **every** point $x \in (a, b)$.

f is **differentiable** on a **closed interval** $[a, b]$ if it is differentiable on the interior (a, b) , and if its **right-hand derivative** at a

$$\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h} = \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a},$$

and its **left-hand derivative** at b

$$\lim_{h \rightarrow 0^-} \frac{f(b+h) - f(b)}{h} = \lim_{x \rightarrow b^-} \frac{f(x) - f(b)}{x - b},$$

exists.

Differentiable Functions

Definition

A function f is **differentiable** if it is differentiable at **every** point $x \in D$ in its **domain**.

Let f be a **differentiable function**. The **derivative** of f is the function

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}.$$

Remark

- (i) The process of finding the derivative is called **differentiation**.
- (ii) The derivative of f can also be denoted as $\frac{df}{dx}(x)$ or $\frac{d}{dx}f(x)$.
- (iii) More precisely, $\frac{d}{dx}$ is called the **differentiation operator**; it acts on differentiable functions.

Example

Show that the function $f(x) = |x|$ is differentiable on $\mathbb{R} \setminus \{0\}$, but has no derivative at $x = 0$.

If $x > 0$, $f(x) = x$, and so

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{x+h-x}{h} = 1.$$

If $x < 0$, $f(x) = -x$, and so

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{-(x+h) - (-x)}{h} = -1.$$

Example

Show that the function $f(x) = |x|$ is differentiable on $\mathbb{R} \setminus \{0\}$, but has no derivative at $x = 0$.

However, if $x = 0$,

$$\lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1, \quad \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1,$$

which tells us that $\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$ is undefined.

Remark

This also shows that $f(x) = |x|$ defined on $[0, \infty)$ (or $(-\infty, 0]$) is differentiable.

Example

Show that the function $f(x) = \sqrt{x}$ is differentiable on $x > 0$, but has no derivative at $x = 0$.

$$\begin{aligned} f'(x) &= \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x} = \lim_{z \rightarrow x} \frac{\sqrt{z} - \sqrt{x}}{z - x} \\ &= \lim_{z \rightarrow x} \frac{\sqrt{z} - \sqrt{x}}{(\sqrt{z} - \sqrt{x})(\sqrt{z} + \sqrt{x})} \\ &= \lim_{z \rightarrow x} \frac{1}{\sqrt{z} + \sqrt{x}} \\ &= \frac{1}{2\sqrt{x}} \end{aligned}$$

which is defined for $x > 0$.

Differentiable $\Rightarrow$ Continuous

Theorem

If f is *differentiable* at $x = c$, then f is *continuous* at $x = c$.

Proof.

Aim: Show that given $f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$ exists, then $\lim_{x \rightarrow c} f(x) = f(c)$. Now

$$f(x) = f(c) + (f(x) - f(c)) = f(c) + \frac{f(x) - f(c)}{x - c}(x - c).$$

So,

$$\begin{aligned}\lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} \left[f(c) + \frac{f(x) - f(c)}{x - c}(x - c) \right] = f(c) + \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}(x - c) \\ &= f(c) + \left(\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \right) \cdot \left(\lim_{x \rightarrow c} (x - c) \right) = f(c) + f'(c) \cdot 0 \\ &= f(c).\end{aligned}$$

Basic Differentiation Rules

Constant Functions

Theorem

If f has the constant value $f(x) = c$, then

$$\frac{df}{dx} = \frac{d}{dx}(c) = 0.$$

Proof.

Since $f(x) = c$ is the constant function, so, for any x and h , $f(x + h) = c = f(x)$. So, for any x ,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0.$$



Positive Integer Power Rule

If n is a positive integer, then

$$\frac{d}{dx}x^n = nx^{n-1}.$$

Proof.

Recall that

$$z^n - x^n = (z - x)(z^{n-1} + z^{n-2}x + z^{n-3}x^2 + \cdots + z^2x^{n-3} + zn^{n-2} + x^{n-1}).$$

So, so any x

$$\begin{aligned} f'(x) &= \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x} = \lim_{z \rightarrow x} \frac{z^n - x^n}{z - x} \\ &= \lim_{z \rightarrow x} \frac{(z - x)(z^{n-1} + z^{n-2}x + \cdots + zn^{n-2} + x^{n-1})}{z - x} \\ &= \lim_{z \rightarrow x} (z^{n-1} + z^{n-2}x + \cdots + zn^{n-2} + x^{n-1}) \quad \text{total of } n \text{ terms} \\ &= nx^{n-1}. \end{aligned}$$

Power Rule

More generally, we have the following theorem.

Theorem

For any real number $\alpha \neq 0$,

$$\frac{d}{dx}x^\alpha = \alpha x^{\alpha-1}$$

for all x where the powers x^α and $x^{\alpha-1}$ are defined.

We will establish this result later.

Scalar Multiple Rule

Theorem

If $f(x)$ is a differentiable function of x and c is a constant, then

$$\frac{d}{dx}(cf(x)) = c \frac{d}{dx}f(x).$$

Proof.

By definition,

$$\begin{aligned}\frac{d}{dx}(cf(x)) &= \lim_{h \rightarrow 0} \frac{(cf(x+h)) - (cf(x))}{h} = \lim_{h \rightarrow 0} c \frac{f(x+h) - f(x)}{h} \\ &= c \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{by the scalar multiplication rule of limit} \\ &= c \frac{d}{dx}f(x).\end{aligned}$$



Sum and Difference Rule

Theorem

If $f(x)$ and $g(x)$ are differentiable functions of x , then their sum $(f+g)(x)$ is differentiable, and

$$\frac{d}{dx}(f+g)(x) = \frac{df}{dx}(x) + \frac{dg}{dx}(x).$$

Proof.

$$\begin{aligned}\frac{d}{dx}(f+g)(x) &= \lim_{h \rightarrow 0} \frac{(f+g)(x+h) - (f+g)(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x+h) + g(x+h) - f(x) - g(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{(f(x+h) - f(x)) + (g(x+h) - g(x))}{h} \\&= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \quad \text{by sum rule of limits} \\&= \frac{df}{dx}(x) + \frac{dg}{dx}(x).\end{aligned}$$

Product Rule

Theorem

If $f(x)$ and $g(x)$ are differentiable functions of x , then so is their products $(fg)(x)$, and

$$\frac{d}{dx}(fg)(x) = f(x)\frac{dg}{dx}(x) + g(x)\frac{df}{dx}(x).$$

Proof.

$$\begin{aligned}\frac{d}{dx}(fg)(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h} \\&= \lim_{h \rightarrow 0} f(x+h) \cdot \frac{g(x+h) - g(x)}{h} + \lim_{h \rightarrow 0} g(x) \cdot \frac{f(x+h) - f(x)}{h} \\&= \lim_{h \rightarrow 0} f(x+h) \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} + g(x) \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\&= f(x)\frac{dg}{dx}(x) + g(x)\frac{df}{dx}(x).\end{aligned}$$

Question

The derivative of a product of three differentiable functions

$$\frac{d}{dx}(f \cdot g \cdot h)(x)$$

is

(a)

$$f(x) \frac{d}{dx}(g \cdot h)(x) + g(x) \frac{d}{dx}(f \cdot h)(x) + h(x) \frac{d}{dx}(f \cdot g)(x)$$

(b)

$$f(x)g(x) \frac{d}{dx}h(x) + f(x)h(x) \frac{d}{dx}g(x) + g(x)h(x) \frac{d}{dx}f(x)$$

Answer

We apply the product rule twice:

$$\begin{aligned}\frac{d}{dx}(f \cdot g \cdot h)(x) &= (f \cdot g)(x) \cdot \frac{dh}{dx}(x) + h(x) \cdot \frac{d}{dx}(f \cdot g)(x) \\ &= f(x) \cdot g(x) \cdot \frac{dh}{dx}(x) + h(x) \cdot g(x) \cdot \frac{df}{dx}(x) + h(x) \cdot f(x) \cdot \frac{dg}{dx}(x)\end{aligned}$$

In general,

$$\frac{d}{dx}(f_1 \cdot f_2 \cdots f_n)(x) = \sum_{i=1}^n (f_1 \cdots f_{i-1} \cdot f_{i+1} \cdots f_n)(x) \frac{df_i}{dx}(x).$$

Quotient Rule

Theorem

If $f(x)$ and $g(x)$ are differentiable functions of x with $g(x) \neq 0$, then so the quotient $\left(\frac{f}{g}\right)(x)$ is differentiable at x , and

$$\frac{d}{dx} \left(\frac{f}{g} \right) (x) = \frac{g(x) \frac{df}{dx}(x) - f(x) \frac{dg}{dx}(x)}{g(x)^2}.$$

Proof.

$$\begin{aligned} \frac{d}{dx} \left(\frac{f}{g} \right) (x) &= \lim_{h \rightarrow 0} \frac{\left(\frac{f}{g} \right) (x+h) - \left(\frac{f}{g} \right) (x)}{h} = \lim_{h \rightarrow 0} \frac{g(x)f(x+h) - f(x)g(x+h)}{hg(x+h)g(x)} \\ &= \lim_{h \rightarrow 0} \frac{g(x)f(x+h) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{hg(x+h)g(x)} \\ &= \lim_{h \rightarrow 0} \frac{g(x) \left(\frac{f(x+h) - f(x)}{h} \right) - f(x) \left(\frac{g(x+h) - g(x)}{h} \right)}{g(x+h)g(x)} = \frac{g(x) \frac{df}{dx}(x) - f(x) \frac{dg}{dx}(x)}{g(x)^2}. \end{aligned}$$

Remarks

Corollary (Reciprocal Rule)

Suppose $f(x)$ is a differentiable function of x with $f(x) \neq 0$. Then

$$\frac{d}{dx} \frac{1}{f(x)} = -\frac{f'(x)}{f(x)^2}.$$

The proof of this corollary is left as an exercise.

With the differentiation rules established so far, we can now differentiate **all rational functions**.

Recall that the **power rule** has been proved only for positive integers. Using the reciprocal rule, one can extend the proof to cover negative integers as well. Students are encouraged to supply the full justification.

Chain Rule

Proposition

$$\frac{d}{dx}(f \circ g)(x) = \frac{df}{dx}(g(x)) \frac{dg}{dx}(x).$$

Sketch of Proof.

$$\begin{aligned} \frac{d}{dx}(f \circ g)(x) &= \lim_{z \rightarrow x} \frac{f(g(z)) - f(g(x))}{z - x} = \lim_{z \rightarrow x} \left[\frac{f(g(z)) - f(g(x))}{z - x} \cdot \frac{g(z) - g(x)}{g(z) - g(x)} \right] \\ &= \lim_{z \rightarrow x} \left[\frac{f(g(z)) - f(g(x))}{g(z) - g(x)} \cdot \frac{g(z) - g(x)}{z - x} \right] = \lim_{z \rightarrow x} \frac{f(g(z)) - f(g(x))}{g(z) - g(x)} \cdot \lim_{z \rightarrow x} \frac{g(z) - g(x)}{z - x} \\ &= \frac{df}{dx}(g(x)) \frac{dg}{dx}(x). \end{aligned}$$



Remarks

In Leibniz's notation, if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx},$$

where $\frac{dy}{du}$ is evaluated at $u = g(x)$.

Alternatively, write

$$\frac{d}{dx} f \circ g(x) = f'(g(x))g'(x).$$

Question

Can you remember the following limits?

1. $\lim_{h \rightarrow 0} \frac{\sin(h)}{h}$

2. $\lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h}$

Question

Can you remember the following limits?

1. $\lim_{h \rightarrow 0} \frac{\sin(h)}{h}$

Answer: $\lim_{h \rightarrow 0} \frac{\sin(h)}{h} = 1$

2. $\lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h}$

Answer: $\lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} = 0$

Derivative of Sine

Recall the angle sum identity:

$$\sin(x + h) = \sin(x) \cos(h) + \cos(x) \sin(h).$$

Hence,

$$\begin{aligned} \frac{d}{dx} \sin(x) &= \lim_{h \rightarrow 0} \frac{\sin(x + h) - \sin(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin(x) \cos(h) + \cos(x) \sin(h) - \sin(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin(x)(\cos(h) - 1) + \cos(x) \sin(h)}{h} = \sin(x) \cdot \underbrace{\lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h}}_{\text{limit 0}} + \cos(x) \cdot \underbrace{\lim_{h \rightarrow 0} \frac{\sin(h)}{h}}_{\text{limit 1}} \\ &= \sin(x) \cdot 0 + \cos(x) \cdot 1 = \cos(x). \end{aligned}$$

The derivative of the sine function is the cosine function

$$\frac{d}{dx} \sin(x) = \cos(x).$$

Derivative of Cosine

Recall the angle sum identity:

$$\cos(x + h) = \cos(x) \cos(h) - \sin(x) \sin(h).$$

Hence,

$$\begin{aligned} \frac{d}{dx} \cos(x) &= \lim_{h \rightarrow 0} \frac{\cos(x + h) - \cos(x)}{h} = \lim_{h \rightarrow 0} \frac{\cos(x) \cos(h) - \sin(x) \sin(h) - \cos(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos(x)(\cos(h) - 1) - \sin(x) \sin(h)}{h} = \cos(x) \cdot \underbrace{\lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h}}_{\text{limit 0}} - \sin(x) \cdot \underbrace{\lim_{h \rightarrow 0} \frac{\sin(h)}{h}}_{\text{limit 1}} \\ &= \cos(x) \cdot 0 - \sin(x) \cdot 1 = -\sin(x) \end{aligned}$$

The derivative of the cosine function is the negative of the sine function

$$\frac{d}{dx} \cos(x) = -\sin(x).$$

Derivative of Other Basic Trigonometric Functions

Since $\sin(x)$ and $\cos(x)$ are differentiable function of x , then so are

$$\tan(x) = \frac{\sin(x)}{\cos(x)}, \quad \cot(x) = \frac{\cos(x)}{\sin(x)}, \quad \sec(x) = \frac{1}{\cos(x)}, \quad \text{and} \quad \csc(x) = \frac{1}{\sin(x)}.$$

Exercise. Using the quotient rule, reciprocal rule, and the derivative of sine and cosine function to derive the following.

$$\begin{aligned} \frac{d}{dx} \tan(x) &= \sec^2(x), & \frac{d}{dx} \cot(x) &= -\csc^2(x), \\ \frac{d}{dx} \sec(x) &= \sec(x) \tan(x), & \frac{d}{dx} \csc(x) &= -\csc(x) \cot(x). \end{aligned}$$

Derivative of Exponential Function

We will state, but not prove, the following theorem. The proof will be given later, after we have formally defined the exponential and logarithmic functions.

Theorem

The derivative of the natural exponential function is itself,

$$\frac{d}{dx}e^x = e^x.$$

Example

1. Find $\frac{d}{dt} \cos(t^2 + 1)$.

Let $f(x) = \cos(x)$ and $g(t) = t^2 + 1$. Then $f'(x) = -\sin(x)$, $g'(t) = 2t$, and $\cos(t^2 + 1) = f \circ g(t)$. Hence,

$$\frac{d}{dt} \cos(t^2 + 1) = f'(g(t))g'(t) = -\sin(t^2 + 1)(2t) = -2t \sin(t^2 + 1).$$

2. Find $\frac{d}{dx} e^{\cos(x)}$.

Let $f(x) = e^x$ and $g(x) = \cos(x)$. Then $f'(x) = e^x$, $g'(x) = -\sin(x)$, and $e^{\cos(x)} = f \circ g(x)$. Hence,

$$\frac{d}{dx} e^{\cos(x)} = f'(g(x))g'(x) = e^{\cos(x)}(-\sin(x)) = -\sin(x)e^{\cos(x)}.$$

Example

3. Find $\frac{d}{dx} \sin^2(2x+1)$.

Let $f(x) = x^2$, $g(x) = \sin(x)$, $h(x) = 2x+1$. Then $f'(x) = 2x$, $g'(x) = \cos(x)$, $h'(x) = 2$, and $\sin^2(2x+1) = f \circ g \circ h(x)$. Hence,

$$\begin{aligned} \sin^2(2x+1) &= f'(g \circ h(x))(g \circ h(x))' = f'(g \circ h(x))g'(h(x))h'(x) \\ &= 2(\sin(2x+1))\cos(2x+1)(2) \\ &= 4\sin(2x+1)\cos(2x+1) \end{aligned}$$

Higher Order Derivatives

Let

$$f(x) = x^n.$$

The **derivative** is $f'(x) = nx^{n-1}$. This is also a **differentiable function**, and we are allowed to take its **derivative**,

$$\frac{d^2}{dx^2} f(x) = \frac{d}{dx} f'(x) = n(n-1)x^{n-2}.$$

This is still a **differentiable function**, and we are allowed to take its **derivative**. Continuing this process, after **differentiating k -times**, we have

$$\frac{d^k}{dx^k} f(x) = n(n-1) \cdots (n-k+1)x^{n-k}.$$

Alternative notation is $f^{(k)}(x) = \frac{d^k}{dx^k} f(x)$.

Example

Find

$$\frac{d^k}{dx^k} \sin(x)$$

$$\frac{d}{dx} \sin(x) = \cos(x)$$

$$\frac{d^2}{dx^2} \sin(x) = \frac{d}{dx} \cos(x) = -\sin(x)$$

$$\frac{d^3}{dx^3} \sin(x) = \frac{d}{dx} (-\sin(x)) = -\cos(x)$$

$$\frac{d^4}{dx^4} \sin(x) = \frac{d}{dx} (-\cos(x)) = \sin(x)$$

Hence,

$$\frac{d^k}{dx^k} \sin(x) = \begin{cases} \sin(x) & \text{if } k = 4m, & m \in \mathbb{Z}_{\geq 0}, \\ \cos(x) & \text{if } k = 4m + 1, & m \in \mathbb{Z}_{\geq 0}, \\ -\sin(x) & \text{if } k = 4m + 2, & m \in \mathbb{Z}_{\geq 0}, \\ -\cos(x) & \text{if } k = 4m + 3, & m \in \mathbb{Z}_{\geq 0}. \end{cases}$$

Derivative of Inverse Functions

Example

Find the derivative of

$$\frac{d}{dx} \sin^{-1}(x)$$

By definition,

$$\sin(\sin^{-1}(x)) = x.$$

We will differentiate both side, using the chain rule on the left hand side:

$$\begin{aligned} \frac{d}{dx} \sin(\sin^{-1}(x)) &= \frac{d}{dx} x \\ \cos(\sin^{-1}(x)) \frac{d}{dx} \sin^{-1}(x) &= 1 \end{aligned} \implies \frac{d}{dx} \sin^{-1}(x) = \frac{1}{\cos(\sin^{-1}(x))}.$$

Recall that $\cos(\theta) = \sqrt{1 - \sin(\theta)^2}$. Let $\theta = \sin^{-1}(x) \implies \sin(\theta) = x$. Hence,

$$\cos(\sin^{-1}(x)) = \sqrt{1 - \sin(\sin^{-1}(x))^2} = \sqrt{1 - x^2}.$$

Therefore,

$$\frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1 - x^2}}.$$

Exercise

Find the derivative of the following inverse trigonometric function

1. $\frac{d}{dx} \cos^{-1}(x)$

2. $\frac{d}{dx} \tan^{-1}(x)$

Derivative Rule for Inverse

More generally, by definition,

$$f \circ f^{-1}(x) = x.$$

Differentiating both sides and applying the chain rule,

$$1 = \frac{d}{dx}x = \frac{d}{dx}f(f^{-1}(x)) = f'(f^{-1}(x)) (f^{-1})'(x).$$

$$\implies (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}.$$

For this expression to make sense, we require $f'(f^{-1}(x)) \neq 0$, which means that f' must be nonzero on an interval containing $f^{-1}(x)$.

Derivative Rule for Inverse

Theorem (Derivative Rule for Inverse)

If f' is nonzero on an interval containing $f^{-1}(b)$, then f^{-1} is *differentiable* at b , and

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}.$$

Power Rule for Rational Powers

We will now sketch the proof of the power rule for rational powers.

Recall that $x^{\frac{1}{n}} = \sqrt[n]{x}$ is the inverse of x^n , for n positive integer. Let $f(x) = x^n$ and $f^{-1}(x) = x^{\frac{1}{n}}$. Then by the derivative for inverse,

$$\begin{aligned}\frac{d}{dx}x^{\frac{1}{n}} &= \frac{1}{\frac{df}{dx}(f^{-1}(x))} = \frac{1}{n(f^{-1}(x))^{n-1}} = \frac{1}{n\left(x^{\frac{1}{n}}\right)^{n-1}} = \frac{1}{n}x^{\frac{1-n}{n}} \\ &= \frac{1}{n}x^{\frac{1}{n}-1}.\end{aligned}$$

Thus, by the chain rule we have the power rule for rational powers:

$$\frac{d}{dx}x^{\frac{p}{q}} = \frac{p}{q}x^{\frac{p}{q}-1},$$

for all integers $p, q \in \mathbb{Z}$.

Exercise. Students are encouraged to supply the full justification.

Derivative of Natural Logarithm Function

Recall that $\ln(x)$ is the inverse of the exponential function e^x . We will use $\frac{d}{dx}e^x = e^x$ to derive $\frac{d}{dx}\ln(x)$.

Let $f(x) = e^x$ and $f^{-1}(x) = \ln(x)$. By the derivative for inverse,

$$\begin{aligned}\frac{d}{dx}\ln(x) &= \frac{1}{\frac{df}{dx}(f^{-1}(x))} = \frac{1}{e^{f^{-1}(x)}} = \frac{1}{e^{\ln(x)}} \\ &= \frac{1}{x}\end{aligned}$$

Using chain rule, for any positive function $f(x) > 0$,

$$\frac{d}{dx}\ln(f(x)) = \frac{1}{f(x)} \frac{df}{dx} = \frac{f'(x)}{f(x)}.$$

Derivative of $a^{f(x)}$

Let $a > 0$ $a \neq 1$. Then

$$\begin{aligned}\frac{d}{dx} a^x &= \frac{d}{dx} e^{x \ln(a)} \\ &= e^{x \ln(a)} \frac{d}{dx} (x \ln(a)) && \text{by chain rule} \\ &= e^{x \ln(a)} \cdot \ln(a) && \ln(a) \text{ is a constant} \\ &= a^x \ln(a)\end{aligned}$$

More generally, if $f(x)$ is a differentiable function of x , then $a^{f(x)}$ is differentiable with

$$\frac{d}{dx} a^{f(x)} = a^{f(x)} \ln(a) \cdot \frac{df}{dx}(x)$$

Derivative of $\log_a(f(x))$

By the change of base formula,

$$\begin{aligned}\frac{d}{dx} \log_a(x) &= \frac{d}{dx} \left(\frac{\ln(x)}{\ln(a)} \right) = \frac{1}{\ln(a)} \frac{d}{dx} \ln(x) \quad \text{since } \frac{1}{\ln(a)} \text{ is a constant} \\ &= \frac{1}{\ln(a)} \frac{1}{x} = \frac{1}{x \ln(a)}.\end{aligned}$$

More generally, if $f(x)$ is a differentiable function of x , then $\log_a f(x)$ is differentiable with

$$\log_a f(x) = \frac{f'(x)}{f(x) \ln(a)}.$$

Derivative of Inverse Trigonometric Function

Suppose $f(x)$ is a differentiable function of x . Then the following inverse trigonometric functions are differentiable with derivative

$$1. \frac{d}{dx} \sin^{-1}(f(x)) = \frac{1}{\sqrt{1-f(x)^2}} \frac{df}{dx}(x), \quad |f(x)| < 1$$

$$2. \frac{d}{dx} \cos^{-1}(f(x)) = -\frac{1}{\sqrt{1-f(x)^2}} \frac{df}{dx}(x), \quad |f(x)| < 1$$

$$3. \frac{d}{dx} \tan^{-1}(f(x)) = \frac{1}{1+f(x)^2} \frac{df}{dx}(x)$$

$$4. \frac{d}{dx} \cot^{-1}(f(x)) = -\frac{1}{1+f(x)^2} \frac{df}{dx}(x)$$

$$5. \frac{d}{dx} \sec^{-1}(f(x)) = \frac{1}{|f(x)|\sqrt{f(x)^2-1}} \frac{df}{dx}(x), \quad |f(x)| > 1$$

$$6. \frac{d}{dx} \csc^{-1}(f(x)) = -\frac{1}{|f(x)|\sqrt{f(x)^2-1}} \frac{df}{dx}(x), \quad |f(x)| > 1$$

Extreme Values of Functions

Absolute Maximum and Minimum

Definition

Let f be a function with domain D .

- ▶ A point c is an **absolute maximum** value on D if

$$f(c) \geq f(x), \quad \forall x \in D.$$

- ▶ A point c is an **absolute minimum** value on D if

$$f(c) \leq f(x), \quad \forall x \in D.$$

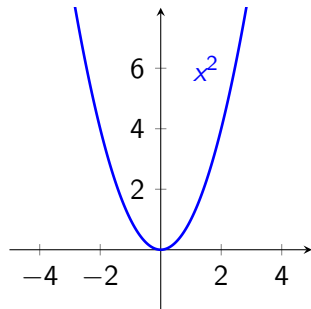
Remark

1. Absolute maximum and minimum values are called **extreme values** of the function f .
2. Absolute maxima and minima are also referred to as **global** maxima or minima.
3. Extreme values **might not exist**, and **might not be unique**.

Example

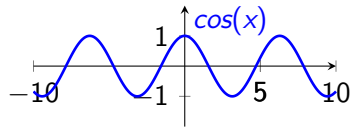
$$f(x) = x^2, \quad D = \mathbb{R}.$$

The point $x = 0$ is an absolute minimum.



$$f(x) = \cos(x), \quad D = \mathbb{R}.$$

The points $x = 2n\pi$, $n \in \mathbb{Z}$ are absolute maximums.



Example

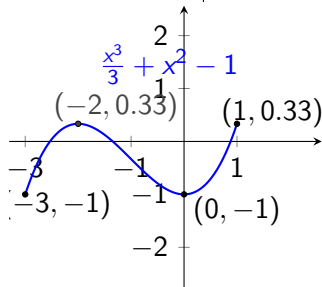
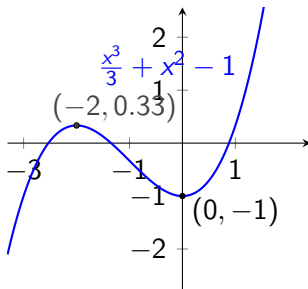
The function

$$f(x) = \frac{x^3}{3} + x^2 - 1$$

has no absolute maximum nor minimum on $D = \mathbb{R}$.

However, if we restrict the domain to $D = [-3, 1]$, then

- ▶ f has absolute maximums at $x = -2$ and $x = 1$,
- ▶ and absolute minimums at $x = -3$ and $x = 0$.



Local Maximum and Minimum

Definition

Let f be a function with domain D .

- ▶ A point c is a **local maximum** value on D if **there exists** and **open interval** $I \subset D$ containing c such that

$$f(c) \geq f(x), \quad \forall x \in I.$$

- ▶ A point c is a **local minimum** value on D if **there exists** and **open interval** $I \subset D$ containing c such that

$$f(c) \leq f(x), \quad \forall x \in I.$$

Remark

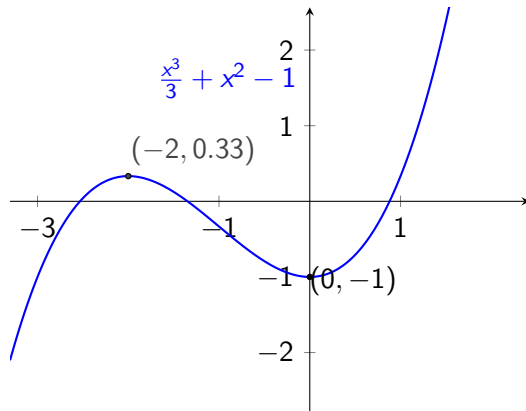
1. Local maximum and minimum values are called **relative extrema** of the function f .
2. Absolute extrema are relative extrema, but functions can have relative but not absolute extrema.
3. Relative extrema **might not exist**, and **might not be unique**.

Example

The function

$$f(x) = \frac{x^3}{3} + x^2 - 1$$

with domain $D = \mathbb{R}$ has local maximum at $x = -2$ and local minimum at $x = 0$.



Local Extrema at Endpoints

Suppose the domain of f is a closed and bounded interval $[a, b]$.

- ▶ Then f has a local maximum at the end point a if there exists a $\delta > 0$ such that

$$f(a) \geq f(x), \quad \forall x \in [a, a + \delta).$$

- ▶ Likewise, f has a local maximum at the end point b if there exists a $\delta > 0$ such that

$$f(b) \geq f(x), \quad \forall x \in (b - \delta, b].$$

- ▶ The inequality is reversed for local minimum values.

Extreme Value Theorem on Closed Bounded Interval

Theorem

Let f be a *continuous* function on a *closed bounded interval* $[a, b]$. Then f attains both an *absolute maximum* value M and an *absolute minimum* value m in $[a, b]$. That is, *there exists* $x_1, x_2 \in [a, b]$ such that

$$f(x_1) = M, \quad f(x_2) = m, \quad \text{and} \quad m \leq f(x) \leq M \quad \forall x \in [a, b].$$

Question

Will the theorem hold if we replace **continuous** with **increasing** or **decreasing** function? That is, is the following statement true.

An **increasing/decreasing** function f on a **closed** and **bounded interval** $[a, b]$ **must attain** an **absolute maximum** and **minimum** value in $[a, b]$.

First Derivative Theorem for Local Extreme Values

Theorem (First Derivative Theorem for Local Extreme Values)

If f has a *local extreme* value at an *interior point* c of its domain, and f is *differentiable* at c , then

$$f'(c) = 0.$$

Example

Let

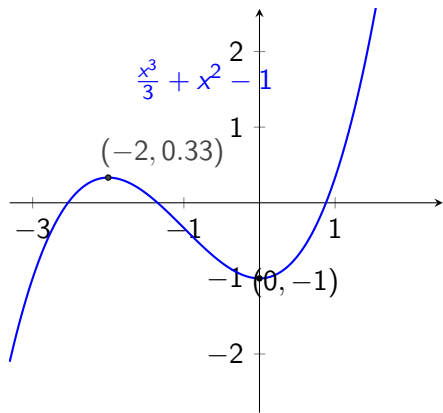
$$f(x) = \frac{x^3}{3} + x^2 - 1.$$

Observe from the graph that f has local extrema at $x = -2$ and $x = 0$. Taking the derivative,

$$f'(x) = x^2 + 2x = x(x + 2)$$

Evaluating at $x = -2$ and $x = 0$,

$$f'(-2) = f'(0) = 0.$$



Question

Is it true that if

$$f'(c) = 0,$$

then $x = c$ is a local extremum of f ?

Question

Is it true that if

$$f'(c) = 0,$$

then $x = c$ is a local extremum of f ?

False. Consider the function

$$f(x) = x^3.$$

$f'(0) = 3(0)^2 = 0$, but $f(0) = 0$ is not a local extrema. The theorem says

$$c \text{ is a local extrema} \Rightarrow f'(c) = 0,$$

not

$$f'(c) = 0 \Rightarrow c \text{ is a local extrema}$$

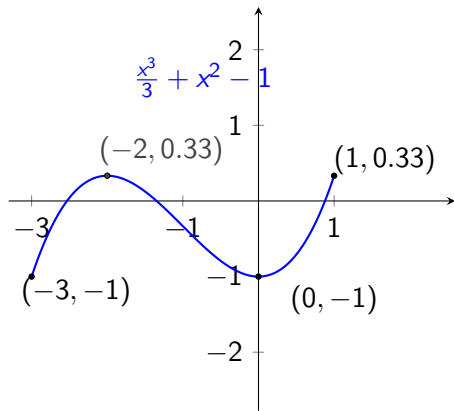
Remark

1. Note that c must be an interior point of the domain. For consider

$$f(x) = \frac{x^3}{3} + x^2 - 1, \quad -3 \leq x \leq 1.$$

$x = -3$ and $x = 1$ are local extrema of f too, but

$$f'(-3) = 3 = f'(1) \neq 0.$$

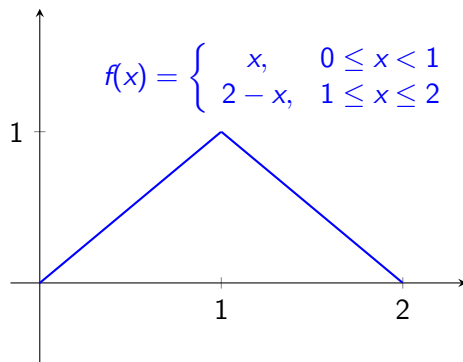


Remark

2. The local extrema of a function could also be at an interior point where f is not differentiable.
For consider the function

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 2 - x, & 1 \leq x \leq 2 \end{cases}$$

$x = 1$ is a local (global) maximum value of f ,
but f' is undefined there.



Critical Points of a Function

Definition

An **interior point** of the **domain** of a function f where f' is **zero** or **undefined** is a **critical point** of f .

Remark

- (i) By the theorem on first derivative for local extrema, every **local extrema** is a **critical point**.
- (ii) The converse is false, that is, not all critical points are local extrema. As seen above, 0 is a critical point of $f(x) = x^3$, but it is not a local extremum. These points are known as **points of inflection** (introduced later).

Finding Absolute Extrema

Finding the Absolute Extrema of a Continuous Function f on a Closed and Bounded Interval

1. Find all critical points of f on the interval.
2. Evaluate f at all critical points and endpoints.
3. Take the largest and smallest of these values.

Example

Find the absolute maximum and minimum values of

$$f(x) = 10x(2 - \ln(x)), \quad x \in [1, e^2].$$

1. $f'(x) = 10(2 - \ln(x)) + 10x(-\frac{1}{x}) = 10(1 - \ln(x))$. This is defined for all $x \in [1, e^2]$.

$$f'(x) = 0 \Leftrightarrow \ln(x) = 1 \Leftrightarrow x = e.$$

There is only one critical point $x = e$.

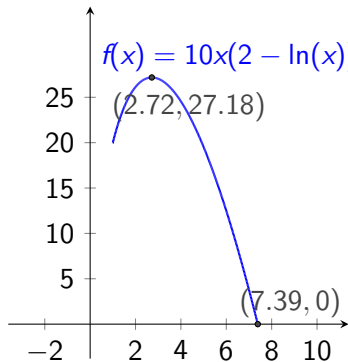
2. The end points are $x = 1$ and $x = e^2$.

Critical point $x = e^1$: $f(e) = 10e(2 - \ln(e)) = 10e$.

Left endpoint $x = 1$: $f(1) = 10(2 - \ln(1)) = 20$.

Right endpoint $x = e^2$: $f(e^2) = 10e^2(2 - \ln(e^2)) = 0$.

3. The absolute maximum value is $x = e$.
The absolute minimum value is $x = e^2$.



Example

Find the absolute maximum and minimum values of

$$f(x) = x^{\frac{2}{3}}, \quad x \in [-2, 3].$$

1.

$$f'(x) = \frac{2}{3}x^{-\frac{1}{3}}, \quad x \neq 0$$

$f'(x)$ is never zero, but is undefined at $x = 0$. So, $x = 0$ is the only critical point.

2. The end points are $x = -2$ and $x = 3$.

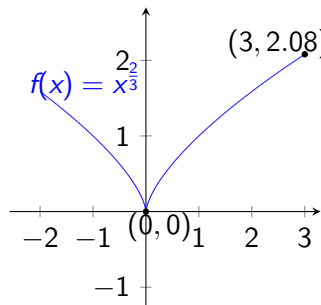
Critical point $x = 0$: $f(0) = 0$.

Left endpoint $x = -2$: $f(-2) = \sqrt[3]{4}$.

Right endpoint $x = 3$: $f(3) = \sqrt[3]{9}$.

3. The absolute maximum value is $x = 3$.

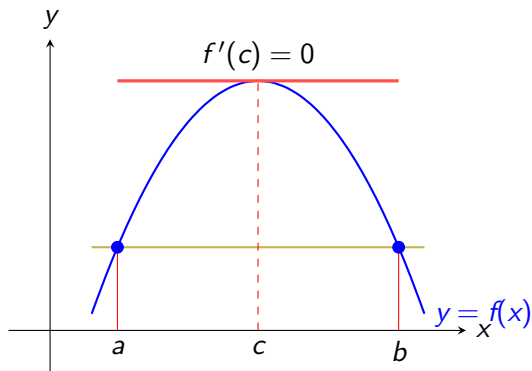
The absolute minimum value is $x = 0$.



Mean Value Theorem

Introduction

Observe from the graph that if a differentiable function crosses a horizontal line at two different points, there is at least one point between them where the tangent to the graph is horizontal and the derivative is zero. This result is known as the Rolle's theorem.



Rolle's Theorem

Theorem

Suppose that $y = f(x)$ is continuous over the closed interval $[a, b]$ and differentiable at every point of its interior (a, b) . If $f(a) = f(b)$, then there is at least one number c in (a, b) at which $f'(c) = 0$.

Proof.

- ▶ Since $f(x)$ is continuous on $[a, b]$, by the extreme value theorem on closed on bounded interval, f must attain both absolute maximum and minimum on $[a, b]$.
- ▶ Since f is differentiable on its interior (a, b) , then either the absolute extrema are interior points where f' is zero, or at a and b .
- ▶ If either of the absolute extrema is at an interior point $c \in (a, b)$, then $f'(c) = 0$, thus proving the theorem.
- ▶ Otherwise, suppose both absolute extrema are at the endpoints. Since $f(a) = f(b)$, it must be the case the f is a constant function. Thus, $f'(x) = 0$ for all $c \in (a, b)$, thus proving the theorem.



Example

Show that the equation

$$f(x) = x^3 + 3x + 1 = 0$$

has exactly one real solution.

Note that $f(x)$ is continuous, $f(-1) = -3$ and $f(0) = 1$. Hence, by the intermediate value theorem, $f(x)$ has a root in the open interval $(-1, 0)$. Suppose now $f(x)$ has 2 roots, say $f(a) = 0 = f(b)$. Then since f is differentiable everywhere, there is a $c \in (a, b)$ such that $f'(c) = 0$. However,

$$f'(x) = 3x^2 + 3$$

which is never zero. So, $f(x)$ has exactly one real root.

Mean Value Theorem

Theorem (Mean Value Theorem)

Suppose f is a *continuous* function defined on a *closed* and *bounded* interval $[a, b]$ and *differentiable* in the *interior* (a, b) . Then *there exists* a point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof.

Let

$$g(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then $g(x)$ satisfies the condition of Rolle's theorem and hence there must be a $c \in (a, b)$ such that

$$0 = g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a} \implies f'(c) = \frac{f(b) - f(a)}{b - a}$$

<https://www.geogebra.org/m/x5t7vn9h>



Remarks

- (i) Suppose $f(x)$ is the distance traveled over time x . Then $\frac{f(b) - f(a)}{b - a}$ is the average speed. Then, at some point c between a and b , the instantaneous speed is the average speed,

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

- (ii) The hypotheses of the theorem are necessary. See remarks on the first derivative theorem for local extrema.
- (iii) This is because the mean value theorem is a corollary of the Rolle's theorem, which in turn is a corollary of the first derivative theorem for local extrema.

Application

Corollary

If $f'(x) = 0$ for all $x \in (a, b)$, then $f(x) = C$ is a constant function over the interval (a, b) .

Proof.

Let $x_1 < x_2$ be any 2 points in (a, b) . Then by the mean value theorem, there is a $c \in (x_1, x_2)$ such that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c) = 0,$$

where the second equality follows from the hypothesis $f'(x) = 0$ for all $x \in (a, b)$. This means that

$$f(x_2) = f(x_1).$$

Since this is true for all $x_1 < x_2 \in (a, b)$, f must be a constant function. □

Application

Corollary

If $f'(x) = g'(x)$ for all $x \in (a, b)$, then there exists a constant $C \in \mathbb{R}$ such that

$$f(x) = g(x) + C \quad \forall x \in (a, b).$$

Proof.

Let

$$h(x) = f(x) - g(x).$$

Then $h'(x) = 0$ for all $x \in (a, b)$. By the previous corollary, $h(x) = C$ is a constant function on (a, b) . So,

$$f(x) - g(x) = h(x) = C \quad \Rightarrow \quad f(x) = g(x) + C.$$



Remark

This will be needed later for indefinite integral/anti-derivative.

Example

Find a function $f(x)$ such that

$$f'(x) = \frac{1}{x}, \quad x > 0$$

and pass through the point $(1, \ln(2))$.

Let $g(x) = \ln(x)$. Then $g'(x) = \frac{1}{x} = f'(x)$ for all $x > 0$, and hence,

$$f(x) = g(x) + C = \ln(x) + C$$

for some constant C . Substituting the point $(1, \ln(2))$, we have

$$\ln(2) = f(1) = \ln(1) + C = C.$$

Hence,

$$f(x) = \ln(x) + \ln(2) = \ln(2x).$$

Example

Suppose f is continuous on $[0, \infty)$ and differentiable on $(0, \infty)$. If

$$f'(x)f(x) = 1 \quad \forall x \in [0, \infty)$$

and $f(0) = 1$. Find $f(x)$.

Recall that $\frac{d}{dx} [f(x)^2] = 2f(x)f'(x)$. So,

$$f'(x)f(x) = 1 \quad \Rightarrow \quad \frac{d}{dx} [f(x)^2] = 2f(x)f'(x) = 2.$$

Consider $g(x) = 2x$, then

$$g'(x) = 2 = \frac{d}{dx} [f(x)^2] \quad \Rightarrow \quad f(x)^2 = g(x) + C = 2x + C$$

for some constant $C \in \mathbb{R}$. Substituting $f(0) = 1$,

$$1^2 = f(0)^2 = 2(0) + C = C \quad \Rightarrow \quad C = 1.$$

Now since $x > 0$, $2x + 1 > 0$ and so,

$$f(x)^2 = 2x + 1 \quad \Rightarrow \quad f(x) = \sqrt{2x + 1}, \quad x > 0.$$

Derivative Test for Local Extrema

Monotone Functions

Corollary

Suppose f is *continuous* on $[a, b]$ and *differentiable* on (a, b) .

- ▶ If $f'(x) > 0$ for all $x \in (a, b)$, then f is *increasing* on $[a, b]$.
- ▶ If $f'(x) < 0$ for all $x \in (a, b)$, then f is *decreasing* on $[a, b]$.

Proof.

We will prove that if $f'(x) > 0$ for all $x \in (a, b)$, then f is increasing. The proof for the second statement is analogous and left as an exercise.

Given any $x_1 < x_2 \in (a, b)$, by the mean value theorem, there is a $c \in (x_1, x_2)$ such that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c) > 0 \quad \Rightarrow \quad f(x_2) - f(x_1) = f'(c)(x_2 - x_1) > 0 \quad \Rightarrow \quad f(x_2) > f(x_1),$$

where the second inequality follows from the fact that $x_2 > x_1$. This proves that f is increasing on $[a, b]$. □

Example

Let

$$p(x) = x^3 - 12x - 5.$$

Find the intervals on which p is increasing and on which p is decreasing.

Note that since p is differentiable everywhere, at the point $x = c$ where f changes from increasing to decreasing, and vice versa, $p'(c) = 0$. So, first find the critical points of p .

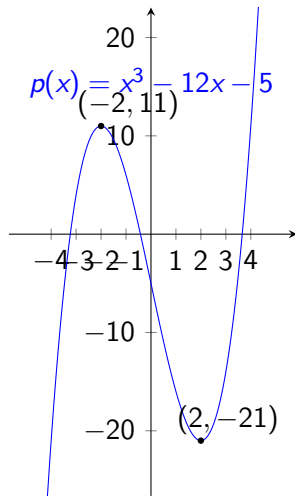
$$p'(x) = 3x^2 - 12 = 0 \quad \Leftrightarrow \quad x = \pm 2.$$

Now for any c in $(-\infty, -2)$ or $(2, \infty)$, $c^2 > 4$, and so

$$p'(c) = 3c^2 - 12 > 0 \quad \Rightarrow \quad p \text{ increasing on } (-\infty, -2) \text{ and } (2, \infty).$$

For any $c \in (-2, 2)$, $c^2 < 4$, and so

$$p'(c) = 3c^2 - 12 < 0 \quad \Rightarrow \quad p \text{ decreasing on } (-2, 2).$$



First Derivative Test for Local Extrema

Proposition (First Derivative Test for Local Extrema)

Suppose

- (a) c is a **critical point** of a **continuous** function f ,
- (b) there is an open interval I containing c such that f is **differentiable** on the puncture interval $I \setminus \{c\}$.

Moving across c from left to right,

1. if f' changes from **negative to positive** at c , then f has a **local minimum** at c ;
2. if f' changes from **positive to negative** at c , then f has a **local maximum** at c ;
3. if f' **does not change sign** at c , then c is **not** a **local extremum** of f .

Idea of Proof.

<https://www.geogebra.org/m/kcrazfx9>



First Derivative Test for Local Extrema

Proof.

We will only proof statement 1. The proof for statement 2 and 3 is analogous, and are left as an exercise.

Since the sign of f' changes from negative to positive at c , there are numbers a and b such that

$$a < c < b, \quad f' < 0 \text{ on } (a, c), \quad \text{and} \quad f' > 0 \text{ on } (c, b).$$

If $x \in (a, c)$, then $f(c) < f(x)$ because $f' < 0$ implies that f is decreasing on $[a, c]$. If $x \in (c, b)$, then $f(c) < f(x)$ because $f' > 0$ implies that f is increasing on $[c, b]$. Therefore, $f(x) \geq f(c)$ for every $x \in (a, b)$. By definition, f has a local minimum at c .



Example

Find the local maximum and local minimum values of

$$f(x) = x^{\frac{2}{3}} e^x,$$

$$f'(x) = \frac{2}{3} x^{-\frac{1}{3}} e^x + x^{\frac{2}{3}} e^x = e^x \left(\frac{2 + 3x}{3x^{1/3}} \right).$$

which is undefined when $x = 0$, and

$$f'(x) = 0 \quad \Leftrightarrow \quad \frac{2}{3} x^{-\frac{1}{3}} = -x^{\frac{2}{3}} \quad \Leftrightarrow \quad x = -\frac{2}{3}$$

Note that $e^x > 0$ for all $x \in \mathbb{R}$. So, for $x < -\frac{2}{3}$, $2 + 3x < 0$ and $x^{1/3} < 0$. Similarly, for $x > 0$, $2 + 3x > 0$ and $x^{\frac{1}{3}} > 0$, and hence,

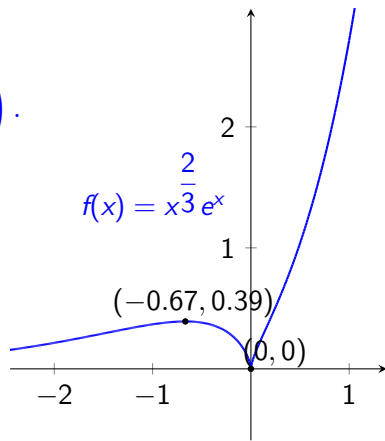
$$f'(x) = e^x \left(\frac{2 + 3x}{3x^{\frac{1}{3}}} \right) > 0 \quad \Rightarrow \quad f \text{ is increasing on } \left(-\infty, -\frac{2}{3} \right) \text{ and } (0, \infty).$$

Example (Continue)

For $-\frac{2}{3} < x < 0$, $2 + 3x > 0$ and $x^{\frac{1}{3}} < 0$, and so,

$$f'(x) = e^x \left(\frac{2 + 3x}{3x^{\frac{1}{3}}} \right) < 0 \quad \Rightarrow \quad f \text{ is decreasing on } \left(-\frac{2}{3}, 0 \right).$$

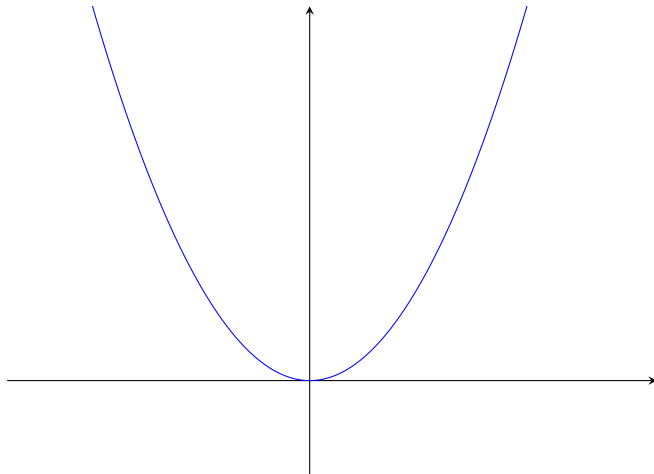
- ▶ Since f' changes from positive to negative at $x = -\frac{2}{3}$, it is a local maximum value.
- ▶ Since f' changes from negative to positive at $x = 0$, it is a local minimum value.



Concavity

The graph of a parabola $y = x^2$ is concave up over any open interval (a, b) .

<https://www.geogebra.org/m/wutqb7mv>



Definition

Definition (Working Definition)

The graph of a differentiable function $y = f(x)$ is

- (a) concave up on an open interval I if f' is increasing on I ;
- (b) concave down on an open interval I if f' is decreasing on I .

Second Derivative Test for Concavity

Proposition

Let $f(x)$ be a *twice-differentiable* function defined on an interval I .

1. If $f'' > 0$ on I , the graph of f is *concave up* on I .
2. If $f'' < 0$ on I , the graph of f is *concave down* on I .

Proof.

Since $f'' > 0$ ($f'' < 0$, respectively), then f' is increasing (decreasing, respectively) on I . Hence, by definition, f is concave up (down, respectively) on I . □

Example

Any quadratic polynomial

$$p(x) = ax^2 + bx + c$$

with $a > 0$ ($a < 0$, respectively) is concave up (down, respectively) on $\mathbb{R}$ since the second derivative is

$$p''(x) = 2a.$$

Suppose $a > 0$. The curve of

$$f(x) = ax^3 + bx^2 + cx + d$$

is concave up on $\left(-\frac{b}{3a}, \infty\right)$ and concave down on $\left(-\infty, -\frac{b}{3a}\right)$ since

$$f''(x) = 6ax + 2b.$$

Inflection Point

Definition

A point $(a, f(a))$ where the **concavity** of the graph of f changes is a **point of inflection**.

<https://www.geogebra.org/m/gmrw3xd5>

Remark

- (i) A point of inflection is also called an **inflection point**.
- (ii) The graph crosses its tangent line at the point of inflection.
- (iii) At a point of inflection $(a, f(a))$, either $f''(a) = 0$ or f'' does not exist.

Example

Determine the concavity and find the inflection points of the function

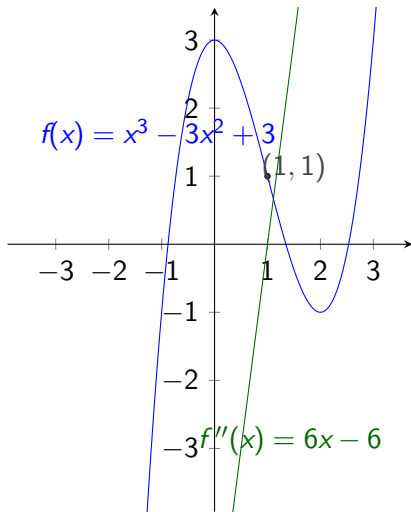
$$f(x) = x^3 - 3x^2 + 3.$$

$$f'(x) = 3x^2 - 6x, \quad f''(x) = 6x - 6.$$

Now $f''(x) > 0$ when $x > 1$, and $f''(x) < 0$ when $x < 1$. So,

- ▶ f is concave down on $(-\infty, 1)$,
- ▶ f is concave up on $(1, \infty)$.

The point of inflection is at $(1, f(1)) = (1, 1)$.



Example

Determine the concavity and find the inflection points of the function

$$f(x) = x^{\frac{5}{3}}.$$

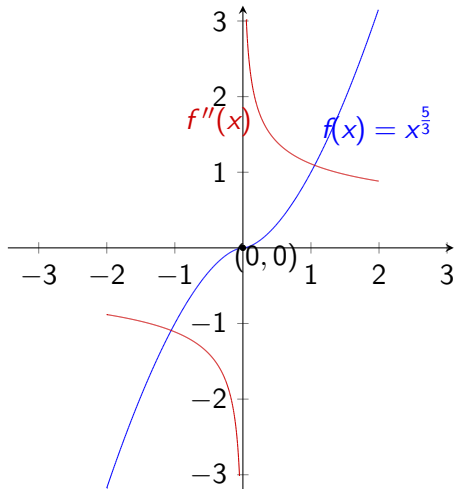
$$f'(x) = \frac{5}{3}x^{\frac{2}{3}}, \quad f''(x) = \frac{10}{9}x^{-\frac{1}{3}}.$$

Now $f''(x) > 0$ when $x > 0$, and $f''(x) < 0$ when $x < 0$. So,

► f is concave down on $(-\infty, 0)$,

► f is concave up on $(0, \infty)$.

f'' is undefined at $x = 0$, but $(0, 0)$ is an inflection point.



Example

Determine the concavity and find the inflection points of the function

$$f(x) = x^{\frac{1}{3}}.$$

$$f'(x) = \frac{1}{3}x^{-\frac{2}{3}}, \quad f''(x) = -\frac{2}{9}x^{-\frac{5}{3}}.$$

► $f''(x) > 0$ when $x < 0$. So, f is concave up on $(-\infty, 0)$.

► $f''(x) < 0$ when $x > 0$. So, f is concave down on $(0, \infty)$.

In this case, f' and f'' is undefined at $(0, 0)$, but it is a point of inflection.

Question

If $f''(a) = 0$ or f'' is undefined at $x = a$, then $(a, f(a))$ is an inflection point. True or False?

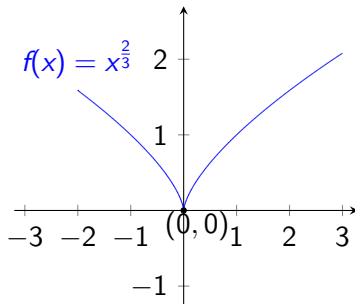
Question

If $f''(a) = 0$ or f'' is undefined at $x = a$, then $(a, f(a))$ is an inflection point. True or False?

False. Consider the function

$$f(x) = x^{\frac{2}{3}}.$$

Then f'' is undefined at $x = 0$. But the concavity does not change at $(0, 0)$; it is not an inflection point.



Second Derivative Test for Local Extrema

Theorem (Second Derivative Test for Local Extrema)

Suppose f'' is continuous on an open interval containing c .

1.

$$f'(c) = 0 \quad \& \quad f''(c) < 0 \quad \Rightarrow \quad f \text{ has a local maximum at } x = c.$$

2.

$$f'(c) = 0 \quad \& \quad f''(c) > 0 \quad \Rightarrow \quad f \text{ has a local minimum at } x = c.$$

Remark

If $f'(c) = 0$ and $f''(c) = 0$, then the test fails. c could be a local maximum value, local minimum value, or neither.

L'Hôpital's Rule

Discussion

Find the limit

$$\lim_{x \rightarrow 0} \frac{2x - \sin(x)}{3x}.$$

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{2x - \sin(x)}{3x} &= \lim_{x \rightarrow 0} \left(\frac{2}{3} - \frac{1}{3} \frac{\sin(x)}{x} \right) \\ &= \frac{2}{3} - \frac{1}{3} \lim_{x \rightarrow 0} \frac{\sin(x)}{x} \\ &= \frac{2}{3} - \frac{1}{3} \\ &= \frac{1}{3}\end{aligned}$$

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{2x - \sin(x)}{3x} &= \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(2x - \sin(x))}{\frac{d}{dx}3x} \\ &= \lim_{x \rightarrow 0} \frac{2 - \cos(x)}{3} \\ &= \frac{2 - 1}{3} \\ &= \frac{1}{3}\end{aligned}$$

Discussion

Find the limit

$$\lim_{x \rightarrow 1} \frac{5x^4 - 4x^2 - 1}{10 - x - 9x^3}.$$

Let $t = x - 1$. When $x \rightarrow 1$, $t \rightarrow 0$.

$$\begin{aligned} & \lim_{x \rightarrow 1} \frac{5x^4 - 4x^2 - 1}{10 - x - 9x^3} \\ = & \lim_{t \rightarrow 0} \frac{5(t+1)^4 - 4(t+1)^2 - 1}{10 - (t+1) - 9(t+1)^3} \\ = & \lim_{t \rightarrow 0} \frac{5t^4 + 20t^3 + 26t^2 + 12t}{-9t^3 - 27t^2 - 28t} \\ = & - \lim_{t \rightarrow 0} \frac{5t^3 + 20t^2 + 26t + 12}{9t^2 + 27t + 28} \\ = & -\frac{3}{7} \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{5x^4 - 4x^2 - 1}{10 - x - 9x^3} &= \lim_{x \rightarrow 1} \frac{\frac{d}{dx}(5x^4 - 4x^2 - 1)}{\frac{d}{dx}(10 - x - 9x^3)} \\ &= \lim_{x \rightarrow 1} \frac{20x^3 - 8x}{-1 - 27x^2} \\ &= \frac{20 - 8}{-1 - 27} \\ &= -\frac{3}{7} \end{aligned}$$

L'Hôpital's Rule

Theorem (L'Hôpital's Rule)

Suppose that f and g are *differentiable* on an *open interval* I containing a point a , $f(a) = g(a) = 0$, and $g'(x) \neq 0$ *for all* $x \in I \setminus \{a\}$. Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

assuming that the limit on the right side of the equation exists.

Example

Find

$$\lim_{x \rightarrow 0} \frac{x - \sin(x)}{x^3}.$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{x - \sin(x)}{x^3} &= \lim_{x \rightarrow 0} \frac{1 - \cos(x)}{3x^2} \quad \text{by L'Hôpital's rule} \\ &= \lim_{x \rightarrow 0} \frac{\sin(x)}{6x} \quad \text{by L'Hôpital's rule} \\ &= \frac{1}{6} \lim_{x \rightarrow 0} \frac{\sin(x)}{x} \\ &= \frac{1}{6}. \end{aligned}$$

Example

Find the limit

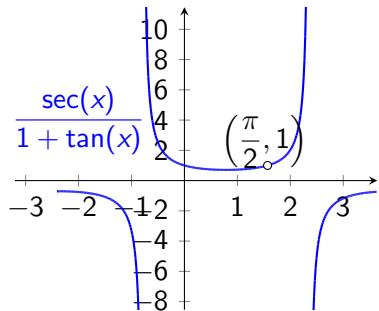
$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec(x)}{1 + \tan(x)}.$$

Substituting $x = \frac{\pi}{2}^-$, we have (written informally)

$$\frac{\sec\left(\frac{\pi}{2}\right)}{1 + \tan\left(\frac{\pi}{2}\right)} = \frac{\infty}{\infty}.$$

Claim: May apply L'Hôpital's rule,

$$\begin{aligned} \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec(x)}{1 + \tan(x)} &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec(x) \tan(x)}{\sec^2(x)} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \sin(x) \\ &= 1. \end{aligned}$$



Example

Find the limit

$$\lim_{x \rightarrow \infty} \frac{\ln(x)}{2\sqrt{x}}.$$

This is of the form $\frac{\infty}{\infty}$. May apply L'Hôpital's rule

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\ln(x)}{2\sqrt{x}} &= \lim_{x \rightarrow \infty} \frac{1/x}{1/\sqrt{x}} = \lim_{x \rightarrow \infty} \frac{\sqrt{x}}{x} \\ &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} \\ &= 0. \end{aligned}$$

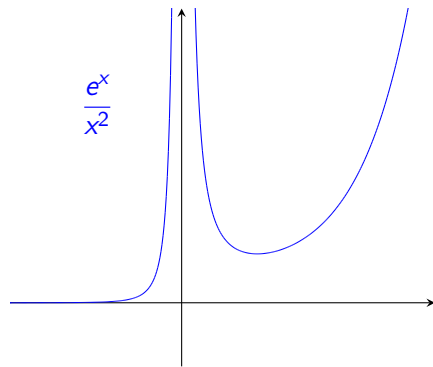
Example

Find the limit

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^2}.$$

This is of the form $\frac{\infty}{\infty}$.

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{e^x}{x^2} &= \lim_{x \rightarrow \infty} \frac{e^x}{2x} \quad \text{still of the form } \frac{\infty}{\infty} \\ &= \lim_{x \rightarrow \infty} \frac{e^x}{2} \\ &= \infty. \end{aligned}$$



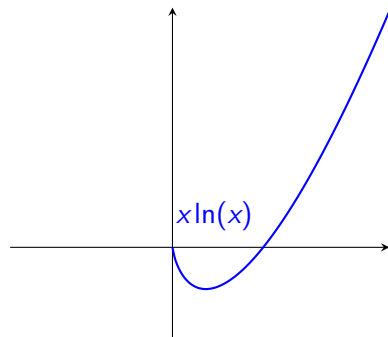
Example

Find the limit

$$\lim_{x \rightarrow 0^+} x \ln(x).$$

This is of the form $0 \cdot (-\infty)$.

$$\begin{aligned} \lim_{x \rightarrow 0^+} x \ln(x) &= \lim_{x \rightarrow 0^+} \frac{\ln(x)}{\frac{1}{x}} \quad \text{this is of the form } \frac{\infty}{\infty} \\ &= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = - \lim_{x \rightarrow 0^+} \frac{x^2}{x} = - \lim_{x \rightarrow 0^+} x \\ &= 0. \end{aligned}$$



Example

Find the limit

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin(x)} - \frac{1}{x} \right).$$

This is of the form $\infty - \infty$.

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1}{\sin(x)} - \frac{1}{x} \right) &= \lim_{x \rightarrow 0} \frac{x - \sin(x)}{x \sin(x)} \quad \text{this is of the form } \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{1 - \cos(x)}{\sin(x) + x \cos(x)} \quad \text{this is of the form } \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{\sin(x)}{\cos(x) + \cos(x) - x \sin(x)} = \lim_{x \rightarrow 0} \frac{\sin(x)}{2 \cos(x) - x \sin(x)} = \frac{0}{2} \\ &= 0. \end{aligned}$$

Question

Find

$$\lim_{x \rightarrow 0} \frac{1 - \cos^2(x)}{x + x^2}.$$

Applying L'Hôpital's rule twice,

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos^2(x)}{x + x^2} &= \lim_{x \rightarrow 0} \frac{2 \sin(x) \cos(x)}{1 + 2x} \\ &= \lim_{x \rightarrow 0} \frac{\sin(2x)}{1 + 2x} \\ &= \lim_{x \rightarrow 0} \frac{2 \cos(2x)}{2} \\ &= 1. \end{aligned}$$

Is this correct?

Summary

1. Use L'Hôpital's rule (repeatedly) whenever finding $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$, where a is either a finite number or $\pm\infty$, and as $x \rightarrow a$,

$$\frac{f(x)}{g(x)} \rightarrow \frac{0}{0}, \frac{\pm\infty}{\pm\infty}.$$

These expressions are known as **indeterminate forms**.

2. May convert the expression of a function $F(x)$ to a form $F(x) = \frac{f(x)}{g(x)}$ such that the above conditions hold for us to use L'Hôpital's rule.
3. Stop using L'Hôpital's rule and evaluate the limit directly if its no longer of indeterminate forms.

Exercises

1. Prove that

$$\lim_{x \rightarrow 0^+} (1+x)^{\frac{1}{x}} = e.$$

Hint: First show that $\lim_{x \rightarrow 0^+} \ln(1+x)^{\frac{1}{x}} = 1$.

2. Find the limit

$$\lim_{x \rightarrow \infty} x^{\frac{1}{x}}.$$

Hint: First find the limit $\lim_{x \rightarrow \infty} \ln \left(x^{\frac{1}{x}} \right)$.

Optimization

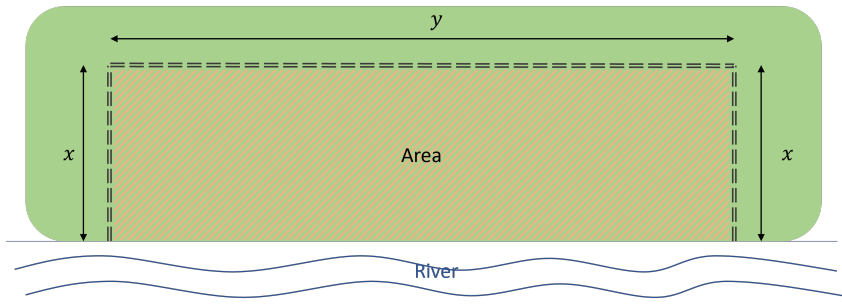
Optimization

Solving applied optimization problems

1. **Read the problem.** Understand the goal of the problem, and identify the hypotheses or assumptions given.
2. **Draw a picture if necessary.** Label the parts that are important to the problem, and include all the assumptions and hypotheses.
3. **Introduce variables.** Identify the variables required to solve the problem, include them in the picture.
4. **Write equations for the unknown variables.** Translate the relationship between the given assumptions and the variables into mathematical equations, make the unknown variables the subject of the formula if necessary.
5. **Test the critical points and endpoints in the domain of the unknown variable.** Use all the tests introduced above to identify and classify the critical points. If the domain is closed and bounded $[a, b]$, remember to check the endpoints.

Area Problem

A farmer has 2400 meters of fencing and wants to fence off a rectangular field that borders a straight river. Suppose he needs no fence along the river. What are the dimensions of the field that has the largest area?



Area Problem

Perimeter: $2400 = 2x + y$, Area: $A = xy$. Aim: Maximize A .

Using $2400 = 2x + y$, we have $y = 2400 - 2x$. Now since $x, y \geq 0$, $0 \leq x \leq 1200$. Hence,

$$\begin{aligned} A &= xy = x(2400 - 2x) = 2400x - 2x^2, 0 \leq x \leq 1200. \\ A'(x) &= 2400 - 4x. \end{aligned}$$

Evaluate A at the critical point $x = 600$ and endpoints $x = 0$ and $x = 1200$.

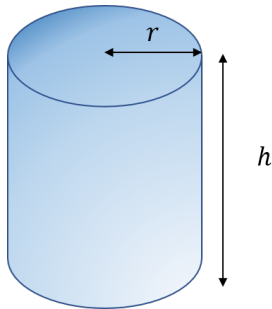
$$\begin{aligned} A(600) &= 2400(600) - 2(600)^2 = 720000, \\ A(0) &= 2400(0) - 2(0)^2 = 0, \\ A(1200) &= 2400(1200) - 2(1200)^2 = 0. \end{aligned}$$

The maximum value is $A(600) = 720000$, at $x = 600$, $y = 1200$.

Conclusion: When the field has dimensions $600 \times 1200\text{m}$, it has the largest area of 720000m^2

Surface Area Problem

A cylindrical can is to be made to hold 1m^3 of oil. Find the dimensions of the can that will minimize the cost of the metal to manufacture the can.



Surface Area Problem

Volume: $1 = \pi r^2 h$. Surface Area: $A = 2\pi r^2 + 2\pi rh$. Aim: Minimize A .

Use $1 = \pi r^2 h$, we have $h = \frac{1}{\pi r^2}$. Now $h, r > 0$. Hence,

$$A = 2\pi r^2 + 2\pi rh = 2\pi r^2 + \frac{2}{r}, r > 0.$$

$$A'(r) = 4\pi r - \frac{2}{r^2} = \frac{2(2\pi r^3 - 1)}{r^2}.$$

The critical point is $r_0 = \frac{1}{\sqrt[3]{2\pi}}$.

$$A''(r) = \frac{4\pi r^3 + 4}{r^3} \Rightarrow A''(r_0) > 0 \Rightarrow r_0 \text{ is a local minimum. OR}$$

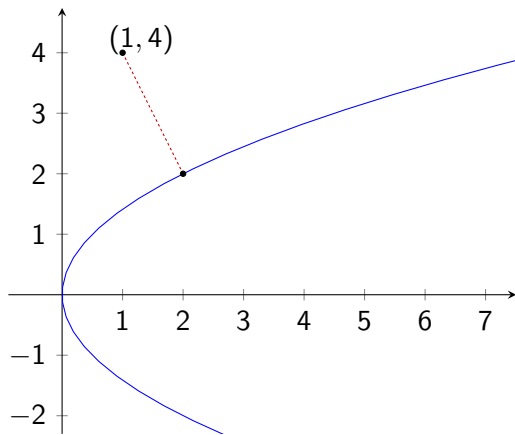
$A'(r) < 0$ for $r \in (0, r_0)$ and $A'(r) > 0$ for $r \in (r_0, \infty) \Rightarrow r_0$ is a local minimum.

The surface area is minimized when $r = \frac{1}{\sqrt[3]{2\pi}}$, $h = \sqrt[3]{\frac{4}{\pi}}$, and the surface area is

$$\left((2\pi)^{\frac{1}{3}} + \frac{2}{\sqrt[3]{2\pi}} \right) \text{ m}^2.$$

Distance Problem

Find the point on $y^2 = 2x$ that is closest to the point $(1, 4)$.



Distance Problem

Curve: $y^2 = 2x$. Distance from (x, y) to $(1, 4)$: $D = \sqrt{(x-1)^2 + (y-4)^2}$. Aim: Minimize D .

By the constrain $y^2 = 2x$, $x > 0$, $y \in \mathbb{R}$.

$$D = \sqrt{\left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2} = \sqrt{\frac{y^4}{4} - 8y + 17}.$$

Observe that minimizing D is equivalent to minimizing $D_2(y) = \frac{y^4}{4} - 8y + 17$.

$$D'_2(y) = y^3 - 8, \quad D''_2(y) = 3y^2$$

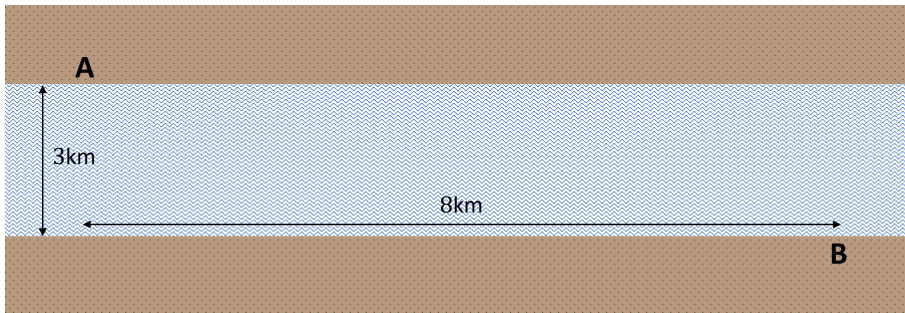
which is always defined, and is 0 if and only if $y = 2$.

$D''_2(2) > 0 \Rightarrow y = 2$ is a local minimum for D .

When $y = 2$, $x = 2$. Hence, $(2, 2)$ is the point on the curve $y^2 = 2x$ that is closest to $(1, 4)$.

Travel Time Problem

A man launches his boat from point A on a bank of a straight river, 3 km wide, and wants to reach point B , 8 km downstream on the opposite bank, as quick as possible. If he can row 6 km/h and run 8 km/h, where should he land?



Travel Time Problem

Let C be the point on the bank directly opposite A , and D be the point the lands. Let x be the distance between C and D . Then the time taken for the man to reach point B is

$$T = \frac{\sqrt{3^2 + x^2}}{6} + \frac{8 - x}{8}, \quad x \in [0, 8].$$

Aim: Minimize T .

$$T'(x) = \frac{x}{6\sqrt{9 + x^2}} - \frac{1}{8}.$$

Evaluate T at the critical point $x = \frac{9}{\sqrt{7}}$ and endpoints.

$$T\left(\frac{9}{\sqrt{7}}\right) = \frac{\sqrt{7}}{8} + 1 \approx 1.331$$

$$T(0) = \frac{3}{2} = 1.5$$

$$T(8) = \frac{\sqrt{73}}{6} \approx 1.424$$

So, the time is minimized when $x = \frac{9}{\sqrt{7}}$